The Impact of Market Composition and Compounding Stressors on Algorithmic Trading Agents

Matthew Poon, psymp18

COMP3004 Designing Intelligent Agents Coursework Report

Abstract

Many algorithmic trading strategies are often evaluated in perfect environments, hiding vulnerabilities exposed by real-world market stressors. This project utilises the Bristol Stock Exchange to simulate Continuous Double Auction markets to evaluate six algorithmic trading strategies. Market simulation runs test each agent's strategies and robustness through varying market compositions and environmental stressors. Results are evaluated against each other through various metrics such as average profit, standard deviation, and t-test scores.

Introduction

Context and Motivation

Algorithmic trading often dominates modern financial exchanges, relying on complex, autonomous agents to execute trades at speeds and volumes beyond human capabilities. The primary mechanism for this is the Continuous Double Auction (CDA): a market structure where multiple buyers and sellers submit bids and asks asynchronously. While many theoretical evaluations of these trading algorithms often assume ideal trading conditions, real-world markets are subject to many different environmental factors.

This project utilises the Bristol Stock Exchange (BSE), an open source simulation of a CDA market created by Dave Cliff (2012), allowing different trading strategies to be tested under competitive market compositions and varying environmental conditions. This makes the topic important as it tests not only whether trading agents can perform well in ideal conditions, but whether their architectures remain reliable when market conditions are disrupted. This is a useful model for AI and autonomy as each agent must perceive a changing environment, choose actions independently, all while competing with other agents.

Research Objectives

This project investigates the following research question: *How does varying market composition affect baseline market efficiency, and to what extent do compounding environmental stressors disrupt this equilibrium across distinct classes of algorithmic traders?*

To answer this question, the project is structured into two distinct phases:

- **Phase 1: Baseline Market Dynamics**
    - Establishes a theoretical baseline by evaluating agent interactions in a clean market environment under varying population compositions.
- **Phase 2: Environmental Ablation Scope**
    - Introduces compounding environmental stressors into the simulation, to determine which architecture offers the greatest resilience in unpredictable environments.

Literature Review & Background

Market Structure and the Continuous Double Auction

The Continuous Double Auction (CDA) allows participants to asynchronously submit limit orders (bids to buy and asks to sell) at any moment, which are recorded in a Limit Order Book (LOB) and executed using price-time priority (O'Hara, 1995). The difference between the highest active bid and the lowest active ask is the spread. Agents attempt to maximise their profitability by providing liquidity within this spread, or by crossing it to take liquidity before prices change.

Algorithmic Trading Architectures

This project evaluates established trading agents using the Bristol Stock Exchange (Cliff, 2012) providing these base architectures: Zero-Intelligence-Constrained (ZIC) traders that generate random quotes constrained only by a predefined budget (Gode and Sunder, 1993), Zero-Intelligence-Plus (ZIP) agents who dynamically adjust internal profit margins based on real-time market feedback (Cliff, 1997), Parameterised-Response Zero-Intelligence (PRZI) agents that use epoch-based optimisation to continuously search a bounded strategy space for optimal pricing parameters (Cliff, 2021), and Shaver (SHVR) agents representing predatory High-Frequency Trading (HFT) by undercutting the current best bid or ask by a single tick. These were selected for this project as they represent varying levels of agent intelligence.

Theoretical Understanding for Custom Agents

Two custom agents extend this: *Momentum* represents trend-following behaviour, incorporating fundamental concepts of Reinforcement Learning to adapt to changing market conditions by evaluating three historical moving averages ($k = 5, 10, 20$). It receives a positive reward ($+1.0$) for matching the actual market movement, and a penalty (-1.0) if it fails. Finally, when generating a new quote, it uses a greedy policy by selecting the highest scoring k-period to calculate the momentum and dynamically scale its bid or ask in that direction.

Additionally, an *Insider* agent was created to establish a theoretical upper bound for market efficiency. It operates with perfect knowledge of the supply and demand schedules, and uses a programmed 10 second front-running mechanism to preemptively dump its inventory before scheduled market crashes to maximise profits.

Environmental Stressors in Financial Markets

Signal noise (imperfect or obscure pricing data) messes up the inputs needed by deterministic algorithms, with Black (1986) noting, “noise” makes financial markets possible but highly imperfect, allowing informed traders to profit from the confusion of noise traders. Execution latency is the delay between data transmission and reception, which hinders HFT algorithms that rely on instantaneous order-book visibility (Hasbrouck and Saar, 2013). And sudden liquidity shocks heavily shift the value of assets, penalising architectures that rely on historical inactivity rather than real-time adaptation.

Methodology & System Design

The Environment

All experiments in this project were conducted using a modified version of the Bristol Stock Exchange (BSE). The core functionality of this CDA market operates a LOB allowing autonomous trading agents to submit bids and asks, and processed via a clearing mechanism. To align with specific objectives of this project, the base architecture was extended to support dynamic environmental stressors, custom agents, and automated batch testing.

Agent Implementations

The simulation evaluated a diverse set of algorithmic traders, consisting of established agents and custom architectures. Standard agents mentioned prior include ZIC, ZIP, PRZI, and SHVR agents.

Momentum was implemented to test whether trend following behaviour is robust when relying on historical prices rather than current market value. It extends the BSE `Trader` superclass, with its `respond()` function overridden to continuously append the latest execution of prices from the market to an internal list (`trade_history`), to ensure computational efficiency while dynamically updating its strategy scores. The `getorder()` function was also adapted to get the current best bid or ask from the LOB dictionary. It then applies its selected momentum delta to this baseline, with

boundary checks `min()` and `max()` to ensure the final quoted price never violates the customer's limit or drops below the system's minimum.

Insider was implemented to provide a privileged benchmark against which the other agents' market efficiency could be compared. It also extended the `Trader` superclass and parsed the `order_schedule` dictionary during initialisation. It extracts price boundaries from the schedule's `ranges` arrays, allowing the agent to calculate the exact pre-crash and post-crash market equilibriums before the simulation starts. Its `getorder()` function uses a 10 second front-running variable (`front_run_lead_time`) to dynamically alter its ordering logic to either return `None` (stop purchasing), or dump its inventory prior to a crash.

These agents were specifically selected to create a meaningful contrast: ZIC provides a random baseline, ZIP tests continuous adaptation, PRZI tests strategy-space optimisation, SHVR tests LOB exploitation, Momentum tests historical trend following, and Insider provides a privileged benchmark. This allows the experiments to compare not just profit values, but different methods of autonomous decision making.

Compounding Environmental Stressors

To truly evaluate algorithmic resilience under simulated real-world conditions, three distinct environmental stressors were added into the BSE system.

First, signal noise was implemented by overriding the base `publish_lob` function. When toggled, a random integer is added directly into the publicly broadcast best bid and ask prices. To maintain system integrity, a boundary check (`max(1, price)`) ensures that injecting noise would never result in negative or zero pricing. This was added as it obscures the true market state, forcing agents to calculate their margins based on corrupted data.

Second, execution latency was implemented within the main `market_session` loop. A random variable is generated against the `latency_level`, which if triggered (`is_lagging = True`), then the agent is bypassed entirely, meaning the system sets `order = None` rather than calling the agent's `getorder()` function. This forces the agent to forfeit for that cycle, thus simulating network packet loss.

Third, market shock was implemented by dynamically splitting the simulation's supply and demand schedules. When enabled, it calculates the simulation midpoint, and at this exact timestamp, the price boundary arrays instantly drop from a base range of *[100, 200]* to *[20, 80]*, representing a sudden market crash, devaluing asset prices. Together, these stressors were implemented to test three different weaknesses in trading agents.

Analytical Dashboard Architecture

In order to manage both the high computational load and data output of the simulation, a custom GUI was created to serve as a control centre and analytical panel, seen in Figure 1. It was implemented using `tkinter`, `pandas`, `matplotlib`, and `scipy`, to merge simulation control, CSV parsing, data visualisation, and statistical testing.

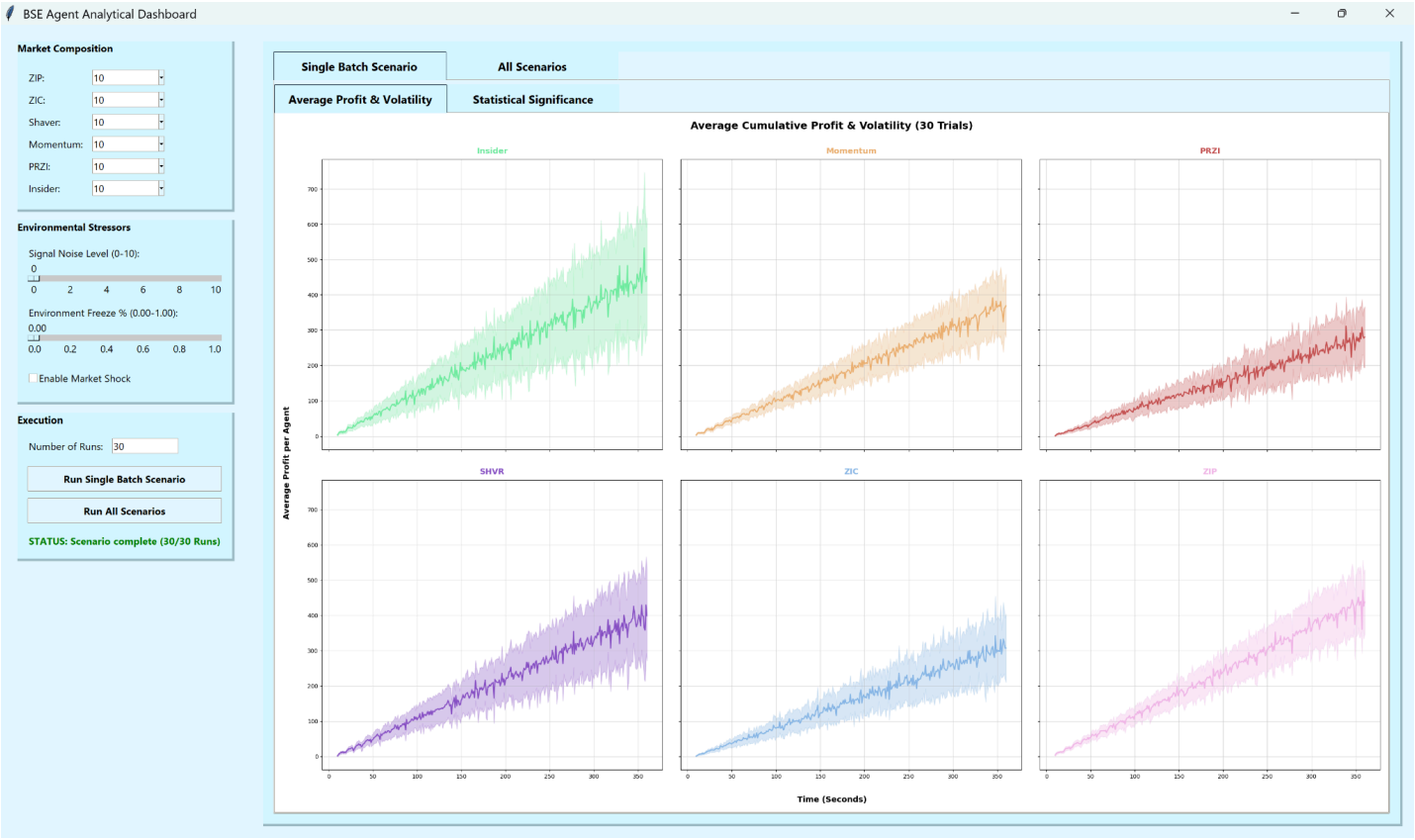


Figure 1. *Full Analytical Dashboard Overview*

The dashboard allows the user to select the number of agents, select stressors to include/omit, and define the number and type of trials, while running BSE in a background thread to prevent the UI from freezing.

To streamline testing, the system parses, aggregates, and calculates the mean and standard deviation across all generated CSV output files after a simulation. These metrics are then mapped to the main visualisation panel.

The dashboard generates plots for the cumulative average profit, standard deviation volatility, T-test matrices, profit degradation charts, and relative market efficiency visualisations.

Experimental Design and Statistical Analysis

To ensure robust statistical testing, every distinct market configuration was simulated across $n = 30$ independent trials with a market runtime of 360 seconds. Performance was evaluated using final mean profit, standard deviation, Welch's T-tests, profit degradation, and relative market efficiency. Welch's T-test was used over Student's T-test as agents often produced unequal variances.

Phase 1 Results and Discussion

The following tests are designed to explore Phase 1 of the project: understanding the effect of varying market composition on different architectures.

Test 1.1: The Balanced Baseline (Establishing the Upper Bound)

Each trading agent is represented equally with ten agents, to demonstrate how each strategy performs under a fair environment.

Rank	Agent	Final Mean Profit	Standard Deviation	Performance Relative to Insider
1	SHVR	£432.47	± 111.70	109.94%
2	Insider	£393.35	± 114.62	100.00%
3	Momentum	£371.81	± 85.08	94.52%
4	ZIP	£342.23	± 77.64	87.00%
5	ZIC	£308.51	± 82.70	78.43%
6	PRZI	£274.70	± 68.92	69.84%

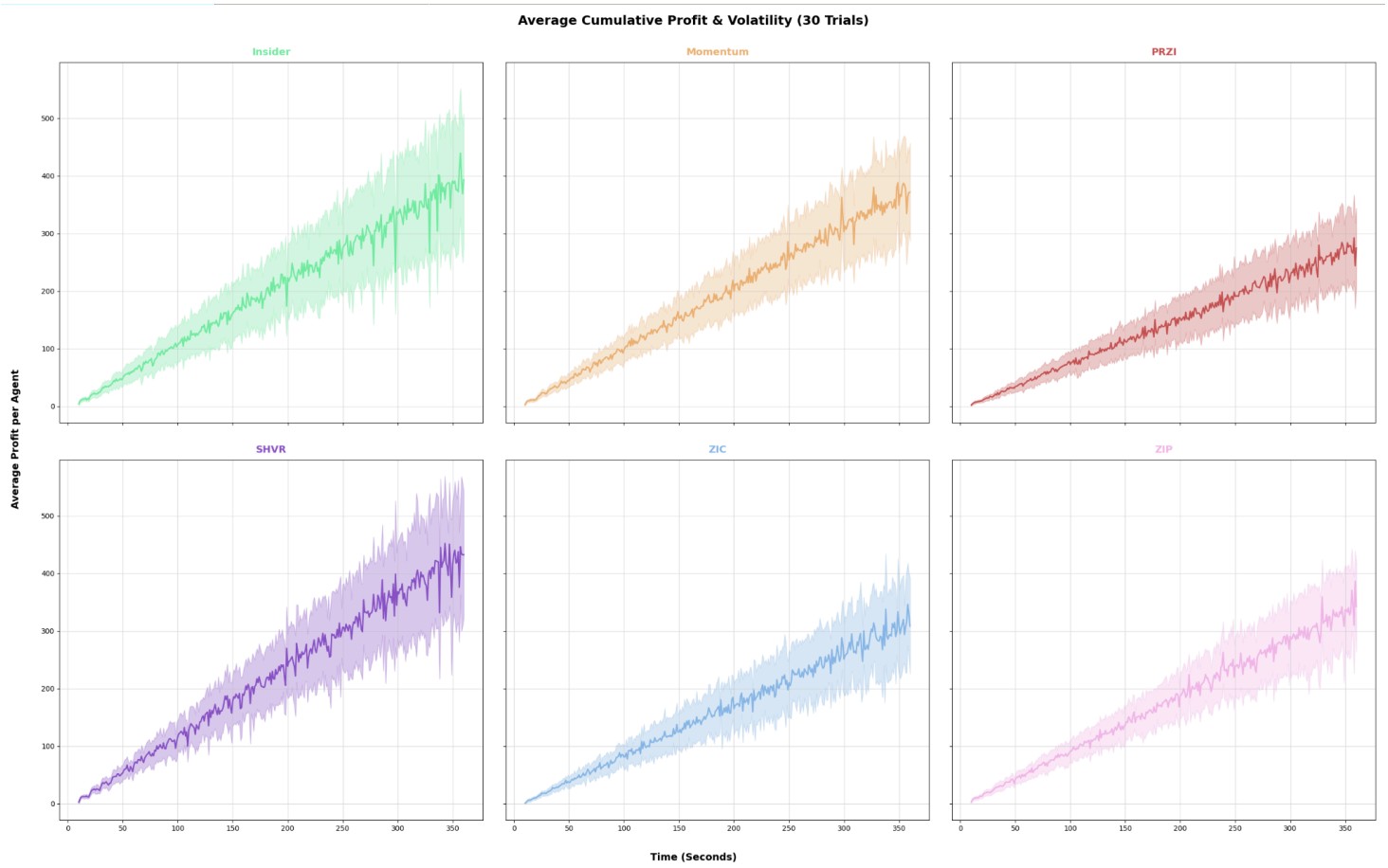


Figure 2. *Test 1.1 Average Profit and Volatility Results*

T-Test Matrix (Statistical Significance at t=360)

Comparison	Agent 1 (Higher)	Agent 2 (Lower)	P-Value	Significant? (p < 0.05)
Rank 1 vs Rank 2	SHVR (432.5)	Insider (393.4)	0.1859	NO
Rank 1 vs Rank 3	SHVR (432.5)	Momentum (371.8)	0.0216	YES
Rank 1 vs Rank 4	SHVR (432.5)	ZIP (342.2)	0.0006	YES
Rank 1 vs Rank 5	SHVR (432.5)	ZIC (308.5)	0.0000	YES
Rank 1 vs Rank 6	SHVR (432.5)	PRZI (274.7)	0.0000	YES
Rank 2 vs Rank 3	Insider (393.4)	Momentum (371.8)	0.4120	NO
Rank 2 vs Rank 4	Insider (393.4)	ZIP (342.2)	0.0484	YES
Rank 2 vs Rank 5	Insider (393.4)	ZIC (308.5)	0.0018	YES
Rank 2 vs Rank 6	Insider (393.4)	PRZI (274.7)	0.0000	YES
Rank 3 vs Rank 4	Momentum (371.8)	ZIP (342.2)	0.1650	NO
Rank 3 vs Rank 5	Momentum (371.8)	ZIC (308.5)	0.0049	YES
Rank 3 vs Rank 6	Momentum (371.8)	PRZI (274.7)	0.0000	YES
Rank 4 vs Rank 5	ZIP (342.2)	ZIC (308.5)	0.1088	NO
Rank 4 vs Rank 6	ZIP (342.2)	PRZI (274.7)	0.0007	YES
Rank 5 vs Rank 6	ZIC (308.5)	PRZI (274.7)	0.0909	NO

Figure 3. *Test 1.1 T-Test Results*

These results show that SHVR surprisingly achieves the highest final mean profit (even above *Insider*) in a balanced, clean market, of £432.47. This suggests that when the LOB is accurate and uninterrupted, SHVR's spread-shaving is highly effective as it can continuously undercut the current best bid/ask by a single tick. The better performance of SHVR over Insider was not statistically significant ($p = 0.1859$), therefore Insider should still be treated as the upper bound benchmark.

Momentum and ZIP also performed strongly, reaching final mean profits of £371.81 and £342.23, respectively. Momentum was not statistically different from Insider and ZIP ($p = 0.4120$ and $p = 0.1650$), suggesting that these agents are highly competitive in clean market conditions.

ZIC and PRZI produced the weakest results, with PRZI achieving the lowest final mean profit (£274.70). As ZIC and PRZI were not significantly different ($p = 0.0909$), PRZI's strategy search did not clearly outperform random quoting in this configuration. Overall, Test 1.1 shows that a balanced, clean market favours agents that can exploit visible LOB structure.

Test 1.2a: Adaptive Intelligence vs Randomness (40 ZIC vs 10 ZIP)

Rank	**Agent**	**Final Mean Profit**	**Standard Deviation**	**Performance Relative to ZIP**
1	ZIP	£466.88	± 109.08	100.00%
2	ZIC	£288.89	± 19.83	61.87%

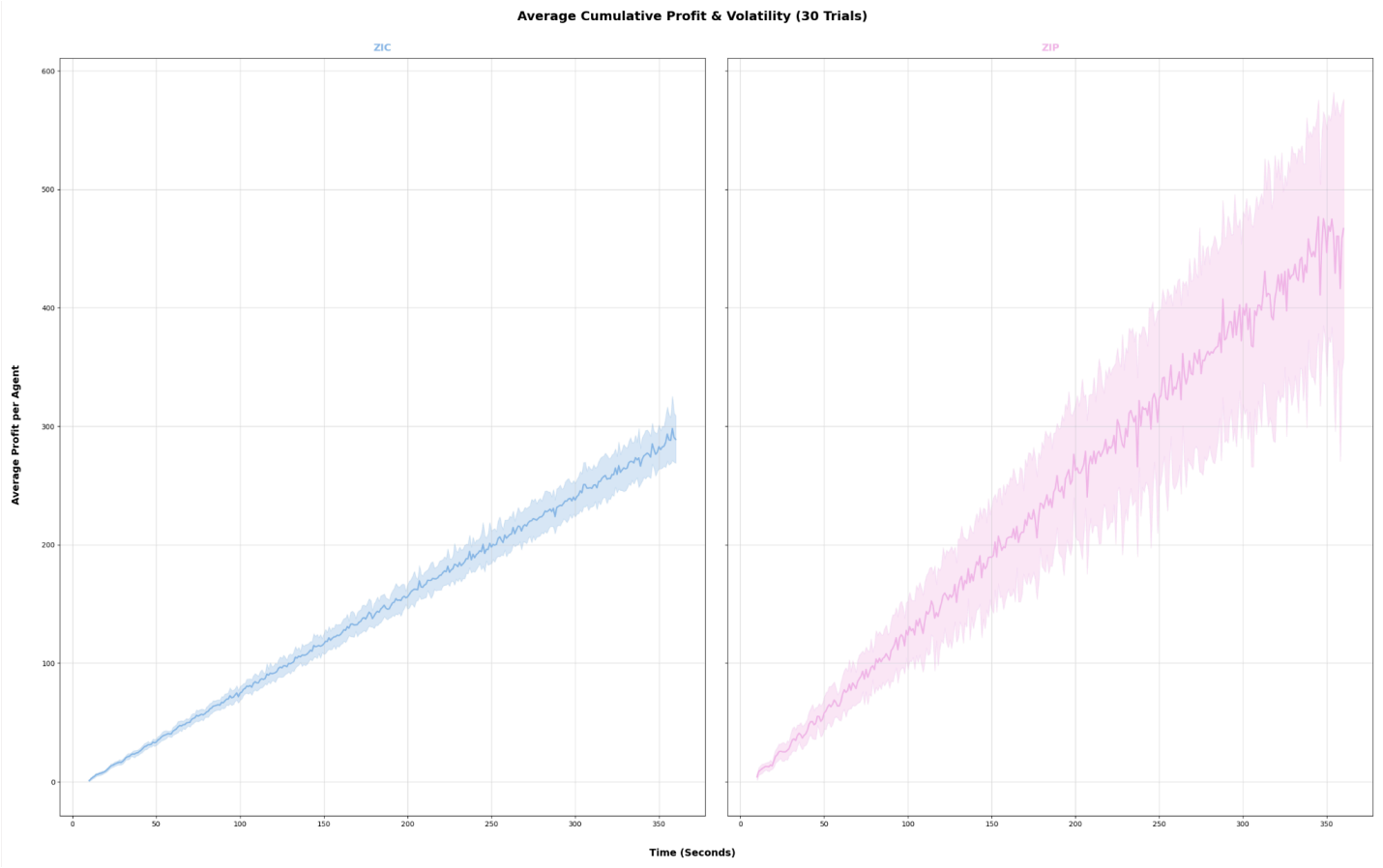


Figure 4. *Test 1.2a Average Profit and Volatility Results*

T-Test Matrix (Statistical Significance at t=360)

Comparison	Agent 1 (Higher)	Agent 2 (Lower)	P-Value	Significant? (p < 0.05)
Rank 1 vs Rank 2	ZIP (466.9)	ZIC (288.9)	0.0000	YES

Figure 5. *Test 1.2a T-Test Results*

These results suggest that in an asymmetrical market, ZIP acts as the dominating architecture when a large population of ZIC agents provide high levels of random liquidity. ZIP's margin-updating adaptability could have taken advantage of ZIC's inefficiencies, to produce a mean final average of £466.88 compared to ZIC's £288.89. ZIC had only reached 61.87% of ZIP's performance, with the T-test results ($p = 0.0000$) confirming that the gap in performance was statistically significant.

Figure 4 indicates ZIC's low volatility (SD ± 19.83), possibly due to the large ZIC population regressing their profits towards the mean (Law of Large Numbers), while ZIP had a much higher volatility (SD ± 109.08) as they may have relied on exploiting the random mistakes of ZIC, therefore performance being bound by the availability of these outliers.

Test 1.2b: High Frequency Spread Manipulation vs Adaptive Intelligence (40 SHVR vs 10 ZIP)

Rank	Agent	Final Mean Profit	Standard Deviation	Performance Relative to SHVR
1	SHVR	£430.74	± 28.56	100.00%
2	ZIP	£385.97	± 97.97	89.61%

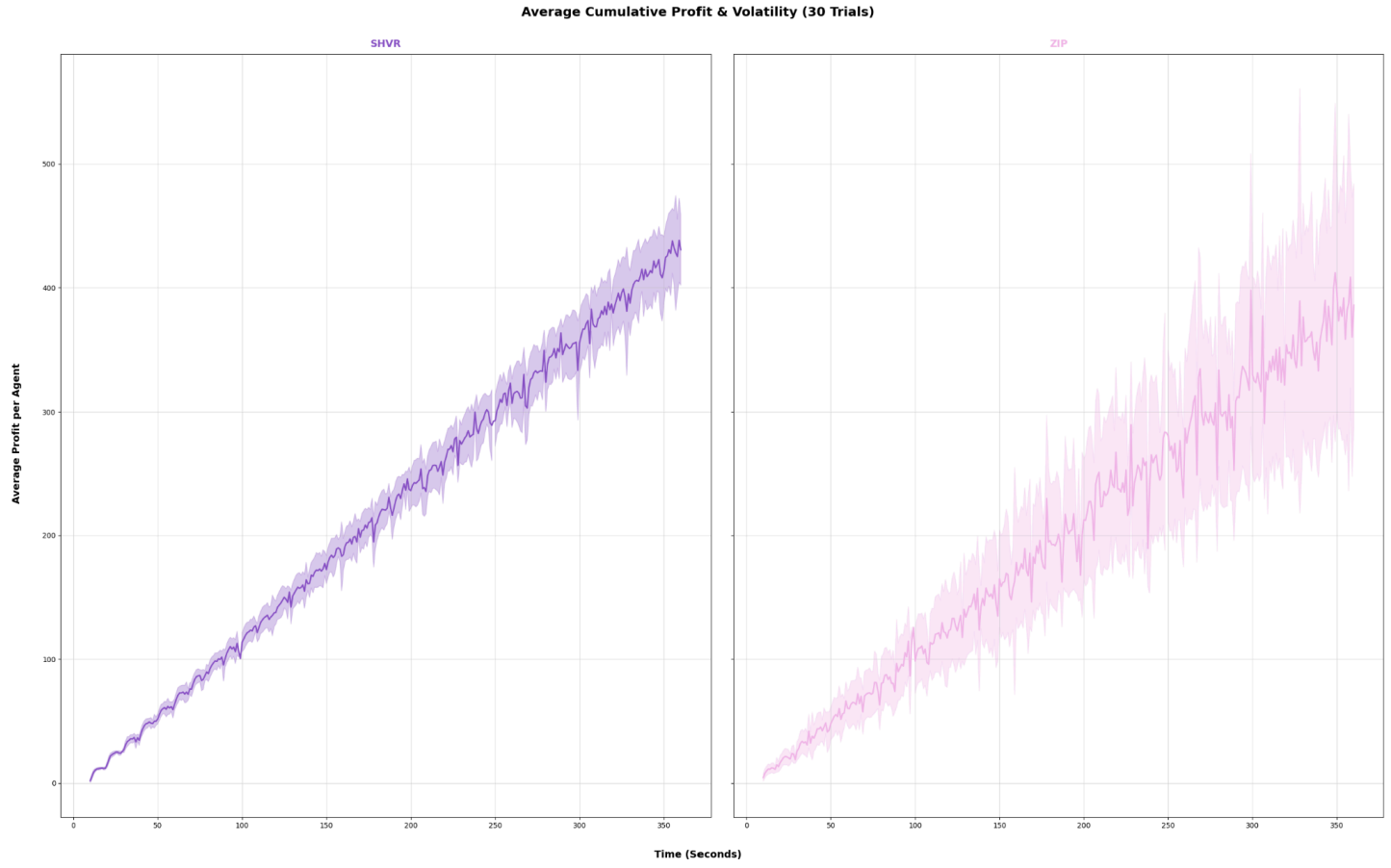


Figure 6. *Test 1.2b Average Profit and Volatility Results*

T-Test Matrix (Statistical Significance at t=360)

Comparison	Agent 1 (Higher)	Agent 2 (Lower)	P-Value	Significant? (p < 0.05)
Rank 1 vs Rank 2	SHVR (430.7)	ZIP (386.0)	0.0217	YES

Figure 7. *Test 1.2b T-Test Results*

These results show SHVR achieved the highest mean final profit in this asymmetrical market, with £430.74 compared to ZIP's £385.97, with the difference being statistically significant ($p = 0.027$), making SHVR's advantage unlikely to be due to random trial variation, and instead through better performance.

Figure 6 supports this, with SHVR's low volatility (SD ± 28.56) suggesting that a large population of spread-shaving agents can produce consistent profit when the LOB is accurate. ZIP also remained profitable, but had a much larger volatility (SD ± 97.97).

This test proposes that although adaptive margin learning agents (ZIP) are highly effective against random constrained traders (ZIC), it struggles to maintain its competitiveness when dominated by aggressive spread-manipulators (SHVR), meaning in this scenario, the speed and volume of SHVR outperformed the adaptive margin learning of ZIP. SHVR is highly effective in clean market conditions when given a population advantage.

Test 1.2c: Trend Following vs Adaptive Intelligence (40 Momentum vs 10 ZIP)

Rank	Agent	Final Mean Profit	Standard Deviation	Performance Relative to ZIP
1	ZIP	£230.47	± 60.20	100.00%
2	Momentum	£94.06	± 17.05	40.81%

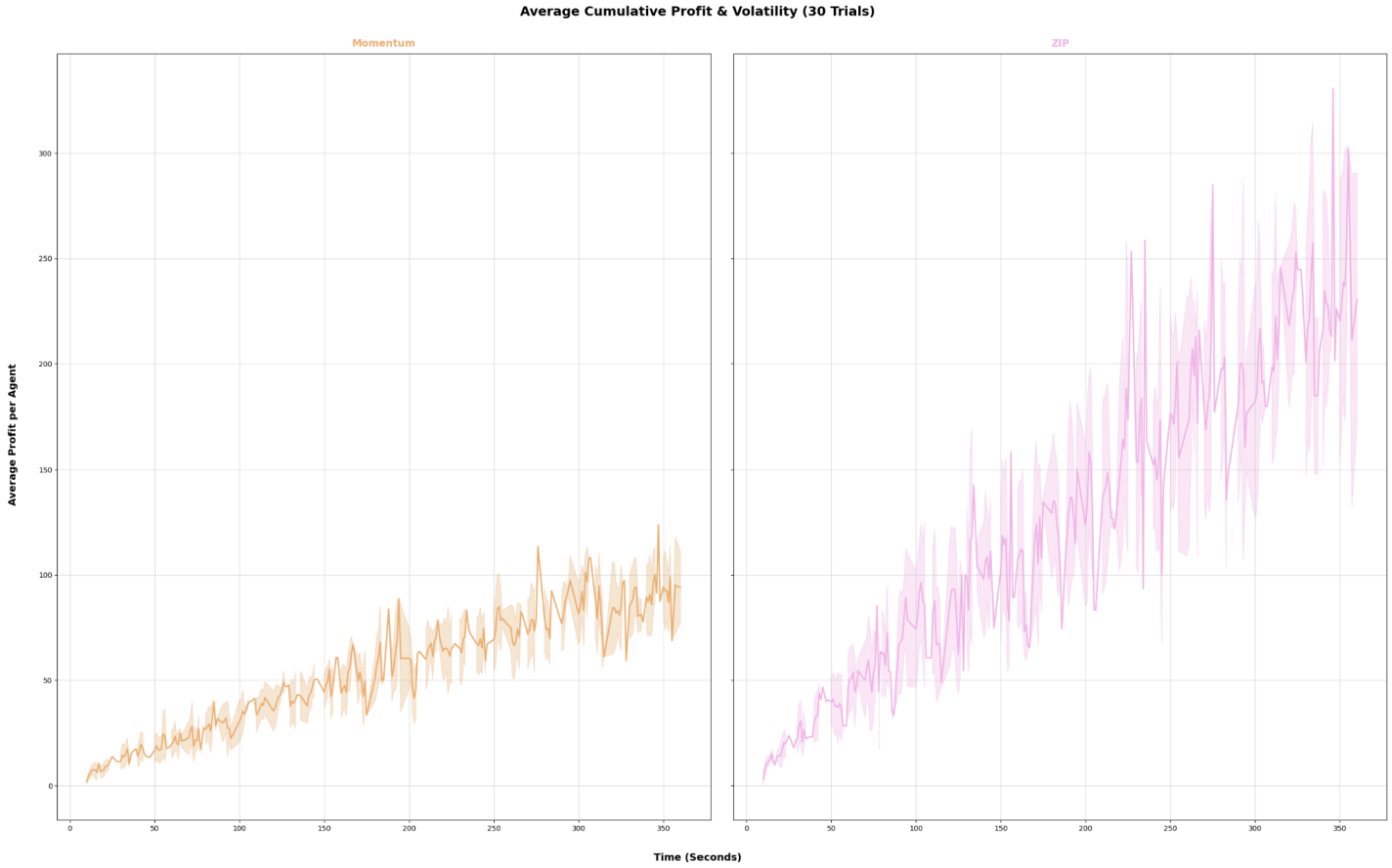


Figure 8 *Test 1.2c Average Profit and Volatility Results*

T-Test Matrix (Statistical Significance at t=360)

Comparison	Agent 1 (Higher)	Agent 2 (Lower)	P-Value	Significant? (p < 0.05)
Rank 1 vs Rank 2	ZIP (230.5)	Momentum (94.1)	0.0000	YES

Figure 9. *Test 1.2c T-Test Results*

This configuration was designed to establish an “echo chamber” structure, evaluating whether a large herd of Momentum agents relying purely on historical moving averages would artificially pump/crash asset prices via a feedback loop. The

results do not support this hypothesis, indicating Momentum did not generate a profitable feedback loop restricting its own final mean profit to £94.06.

Conversely, ZIP again demonstrated great exploitation capabilities, securing a statistically significant advantage ($p = 0.0000$) with a final mean profit of £230.47. As ZIP adapts based on real-time profit margins, they seem to be less affected by Momentum's behaviour.

The standard deviations may reinforce these ideas, as Momentum produced a very low volatility (SD ± 17.05) potentially due to consistent yet limited trend-following behaviour, while ZIP produced a much higher volatility (SD ± 60.20) due to its capability of dynamically scaling its profit margins. Overall, Test 1.2c shows that trend-following was not competitive against margin learning, even with a population advantage.

Test 1.2d: Adaptive Intelligence vs Epoch-Based Optimisation (20 ZIP, 20 PRZI, 10 ZIC)

Rank	Agent	Final Mean Profit	Standard Deviation	Performance Relative to ZIP
1	ZIP	£374.80	± 42.41	100.00%
2	ZIC	£281.71	± 93.58	75.19%
3	PRZI	£266.43	± 44.59	71.08%

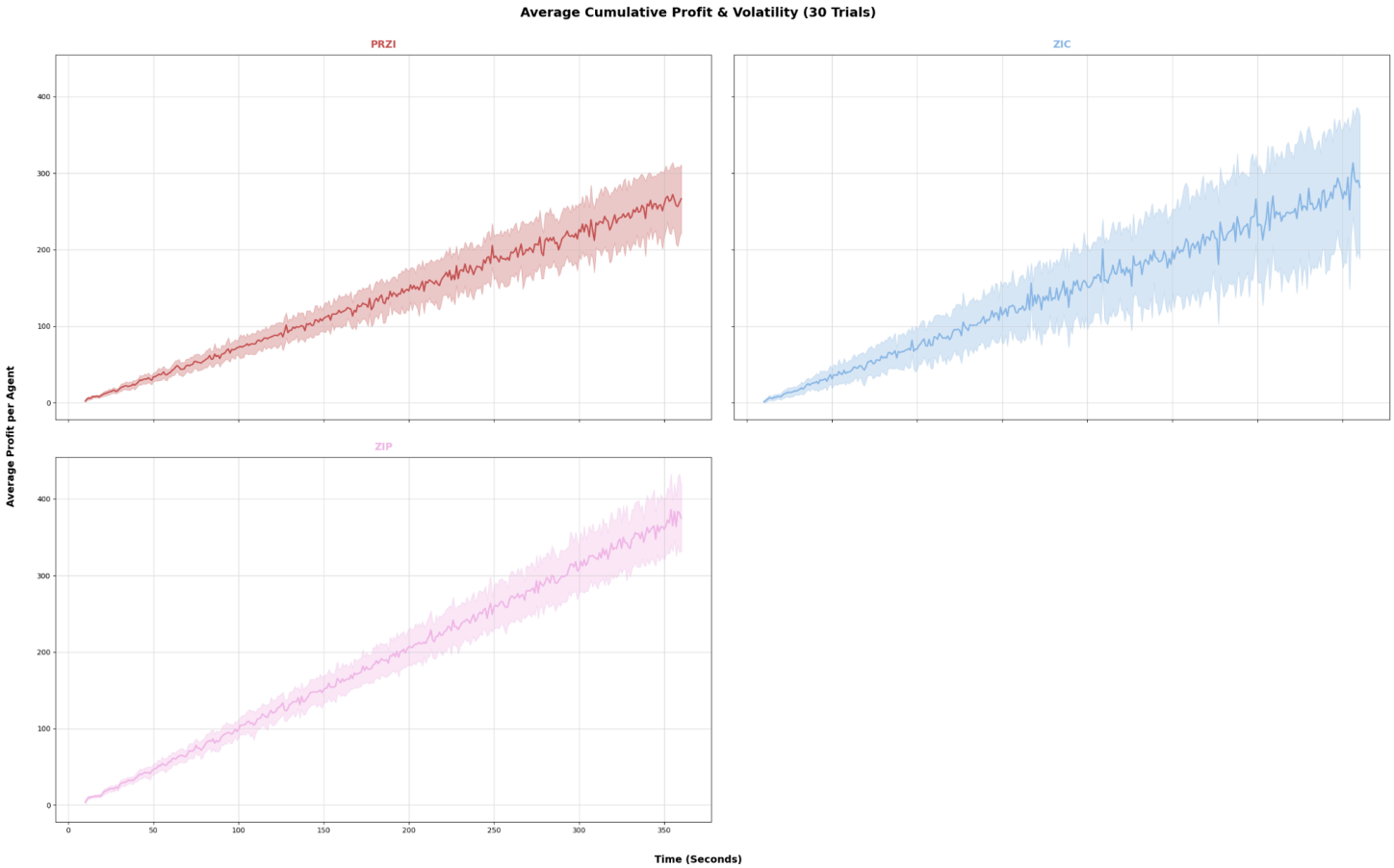


Figure 10. *Test 1.2d Average Profit and Volatility Results*

T-Test Matrix (Statistical Significance at t=360)

Comparison	Agent 1 (Higher)	Agent 2 (Lower)	P-Value	Significant? (p < 0.05)
Rank 1 vs Rank 2	ZIP (374.8)	ZIC (281.7)	0.0000	YES
Rank 1 vs Rank 3	ZIP (374.8)	PRZI (266.4)	0.0000	YES
Rank 2 vs Rank 3	ZIC (281.7)	PRZI (266.4)	0.4240	NO

Figure 11. *Test 1.2d T-Test Results*

The data suggests ZIP's margin-updating heuristic is more effective (£374.80) than PRZI's strategy-space search (£266.43), with T-test results (p = 0.0000) supporting this. ZIP may be able to capture liquidity much faster than PRZI, as ZIP adapts instantaneously whereas PRZI requires longer evaluation periods.

These results reinforce the findings of the baseline test: complex optimisation algorithms are ineffective if their convergence time is too slow. Analysing the variance further reinforces PRZI's limitations, as it produced the lowest volatility (SD ± 44.59) demonstrating that these agents were consistently stuck, unable to adapt fast enough to exploit the random liquidity provided by ZIC.

Although PRZI performed below ZIC's mean final profit (£281.71) this was not statistically significant (p = 0.4240) therefore PRZI should not be interpreted as meaningfully worse than ZIC, but rather as failing to produce a clear advantage in this configuration. This is important as PRZI is more complex than ZIC, yet the results suggest that this complexity did not equate to stronger profitability.

Phase 2 Test Results

The following tests are designed to explore Phase 2 of the project: understanding the effect of varying market conditions (*Control*, *Noise Only*, *Latency Only*, *Shock Only*, and *All Stressors*) and market composition on different architectures.

Test 2.1: All Stressors in a Balanced Population

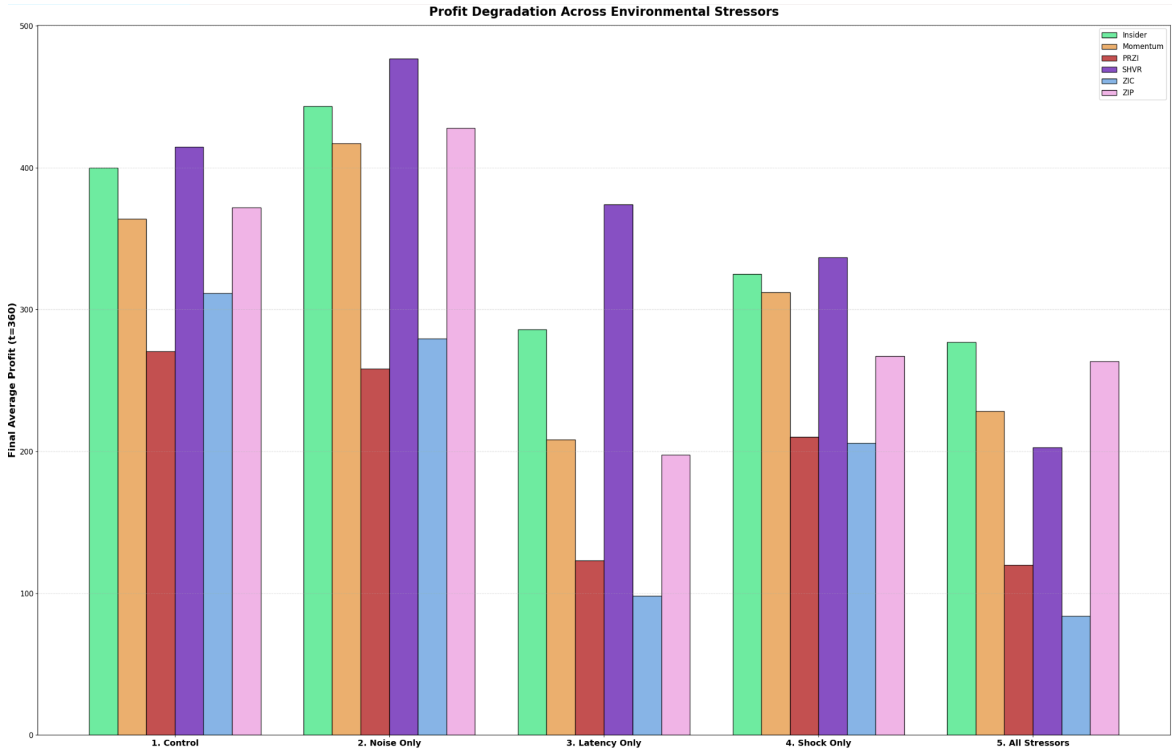


Figure 12. *Test 2.1 Profit Degradation Bar Chart Results*

Agent Efficiency Percentage (Against Insider upper bound)

Agent	1. Control	2. Noise Only	3. Latency Only	4. Shock Only	5. All Stressors
Insider	100.00%	110.82%	71.43%	81.23%	69.21%
Momentum	90.97%	104.30%	52.13%	77.98%	57.10%
PRZI	67.61%	64.54%	30.75%	52.54%	29.96%
SHVR	103.67%	119.21%	93.52%	84.17%	50.72%
ZIC	77.83%	69.82%	24.54%	51.46%	21.00%
ZIP	92.92%	106.95%	49.38%	66.78%	65.79%

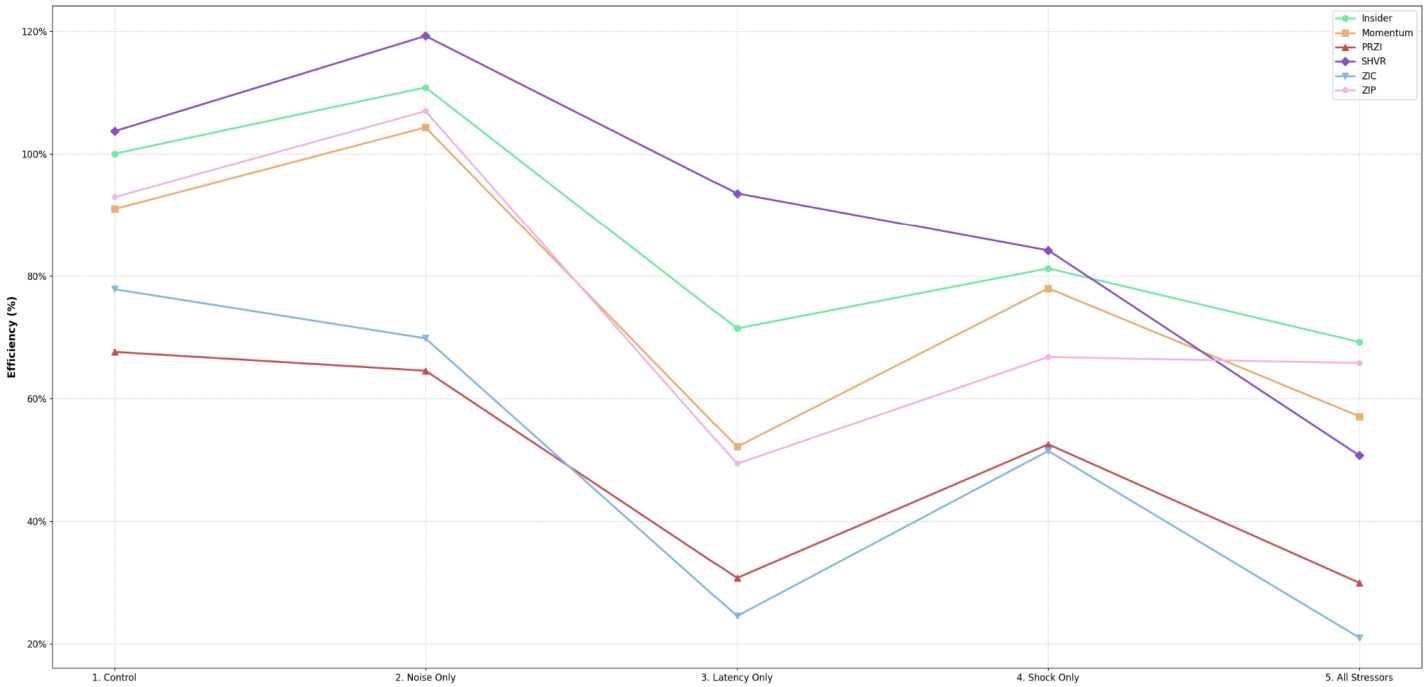


Figure 13. *Test 2.1 Market Efficiency Matrix Results*

These results suggest that environmental stressors significantly impact the profitability from Phase 1's baseline test. In the *Control* scenario, again SHVR surprisingly exceeded *Insider*'s benchmark, reaching 103.67% relative efficiency.

Another surprising finding can be seen in the *Noise Only* scenario. Instead of reducing performance, noise increased the efficiency of several agents: SHVR with 119.21%, Insider with 110.82%, ZIP with 106.85%, and Momentum with 104.30%. This suggests that introducing noise may have enhanced exploitable price inconsistencies instead of hindering performance. Agents that can dynamically react to distorted data appear to have benefited from the added stressor.

Latency Only produced the largest effect, with ZIC falling to 24.54% relative efficiency and PRZI dropping to 30.75%, demonstrating that slower architectures are highly dependent on frequent trading opportunities. Surprisingly, SHVR remained strong under latency at 93.52%, with ZIP and Momentum falling much more heavily with 49.38% and 52.13% efficiency, respectively.

The *Shock Only* scenario also hindered agent performance, but did not collapse the market, with SHVR again remaining the highest at 84.17%, outperforming *Insider* and *Momentum* (81.23% and 77.98%). This could suggest that sudden drops in prices damaged profitability, but allowed agents to recover afterwards. ZIC and PRZI again performed poorly, showing limited resilience to market shifts.

With *All Stressors*, every agent degraded significantly, with SHVR suffering the largest, falling from 119.21% to 50.72% efficiency. ZIP surprisingly performed better under the combined stress over latency alone, reaching 65.79% efficiency,

making it one of the more resilient agents in disrupted market conditions. Overall, Test 2.1 demonstrates that clean market performance does not always correlate with robustness.

Test 2.2a: The Trend Following Breakdown (30 Momentum, 15 ZIC, 5 Insider)

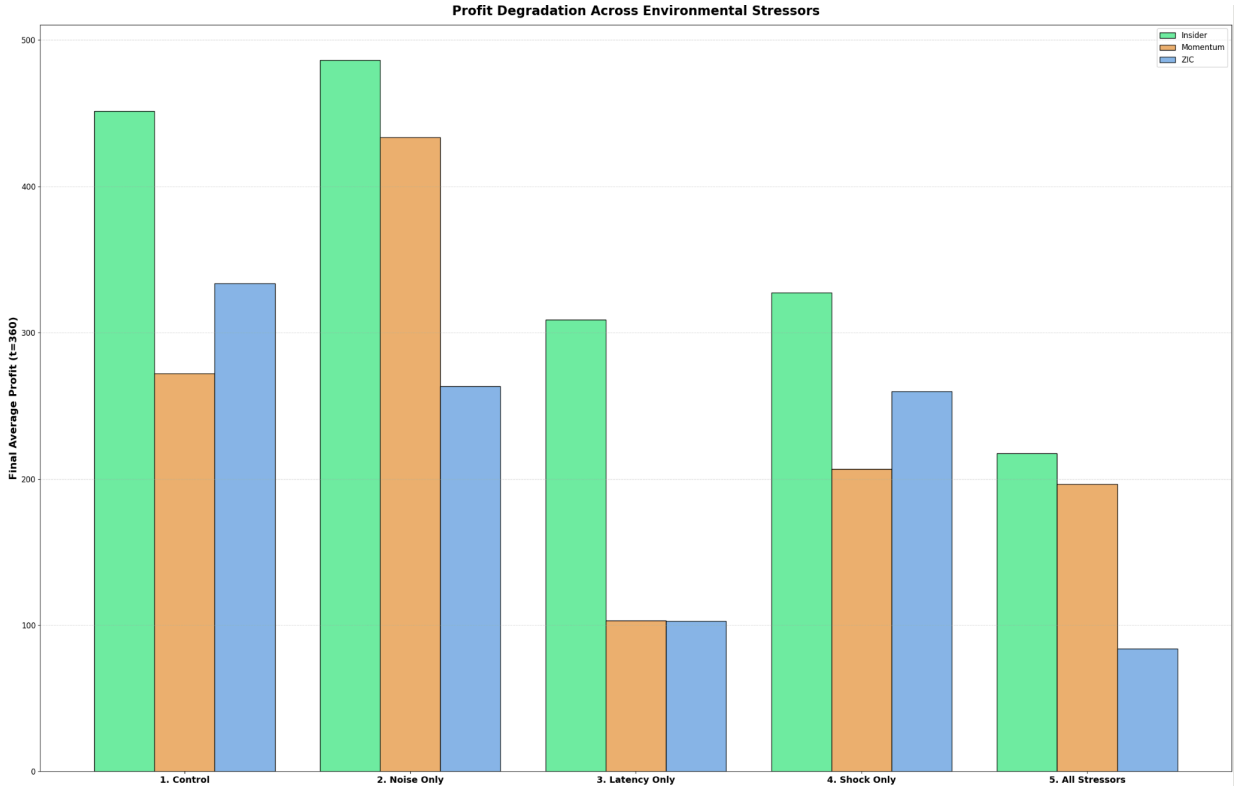


Figure 14. *Test 2.2a Profit Degradation Bar Chart Results*

Agent Efficiency Percentage (Against Insider upper bound)

Agent	1. Control	2. Noise Only	3. Latency Only	4. Shock Only	5. All Stressors
Insider	100.00%	107.69%	68.35%	72.50%	48.17%
Momentum	60.24%	96.00%	22.84%	45.80%	43.55%
ZIC	73.88%	58.29%	22.79%	57.49%	18.60%

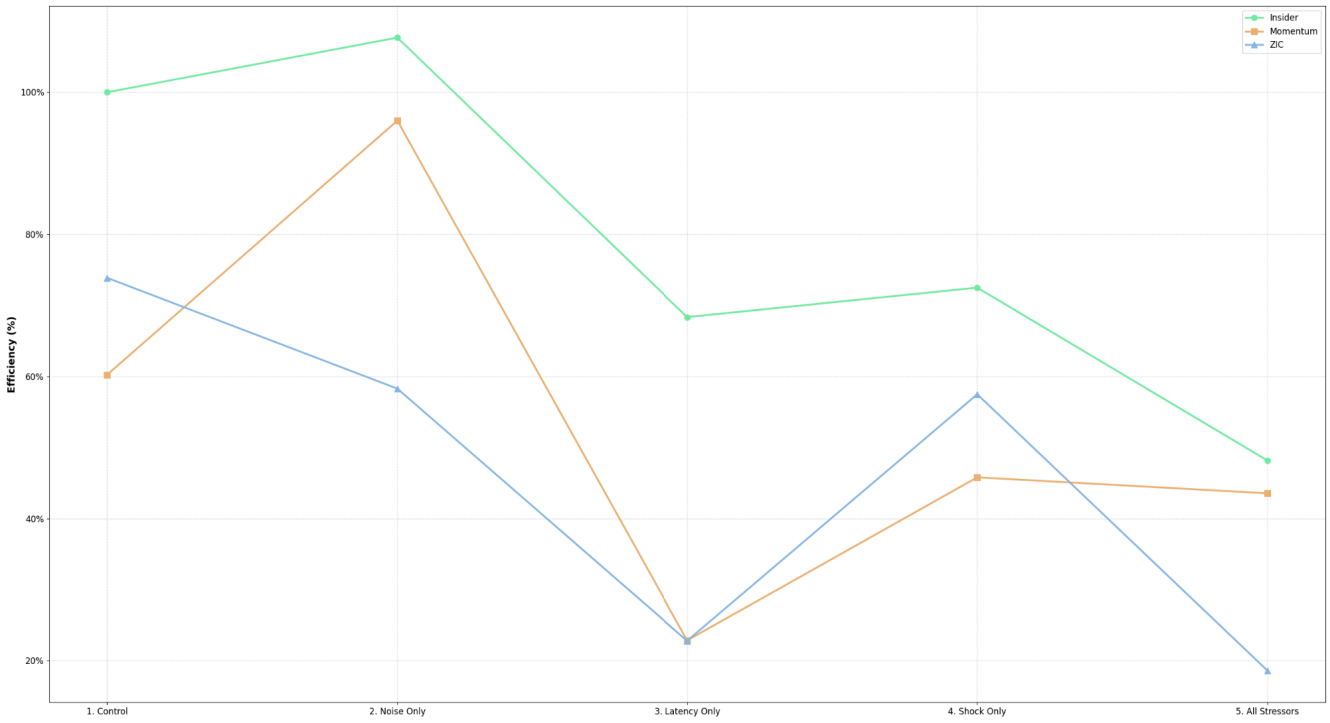


Figure 15. *Test 2.2a Market Efficiency Matrix Results*

In a similar manner to Test 2.1, results show that while the *Control* scenario sets *Insider* as the 100% efficiency benchmark, introducing *Signal Noise* resulted in a surprising efficiency rating boost of 106.58%, and likewise for Momentum (60.24% to 96.00%). However, both the *Shock Only* and *Latency Only* scenario exposes Momentum's structural vulnerability, suffering a 43.55% and 22.84% efficiency drop respectively, as crash would set the moving averages to still indicate assets are highly valued when they are not, directly affecting the reliance on historical prices.

Another interesting observation is that ZIC outperforms Momentum in the *Control* scenario, (73.88% vs 60.24%), as well as the *Shock Only* scenario (57.49% vs 45.80%), and does not fall far behind Momentum's performance with exception to the *Noise Only* scenario and *All Stressors* scenario, performing very poorly at 58.29% efficiency and 18.60% efficiency in these two configurations.

When subjected to *All Stressors* simultaneously, agent performance is hindered significantly, with even *Insider* dropping to 48.17% efficiency, suggesting that perfect pricing knowledge cannot overcome the physical limitations of execution latency and market shocks.

Test 2.2b: High Frequency Spread Manipulation vs Adaptive Intelligence Rematch (40 SHVR vs 10 ZIP)

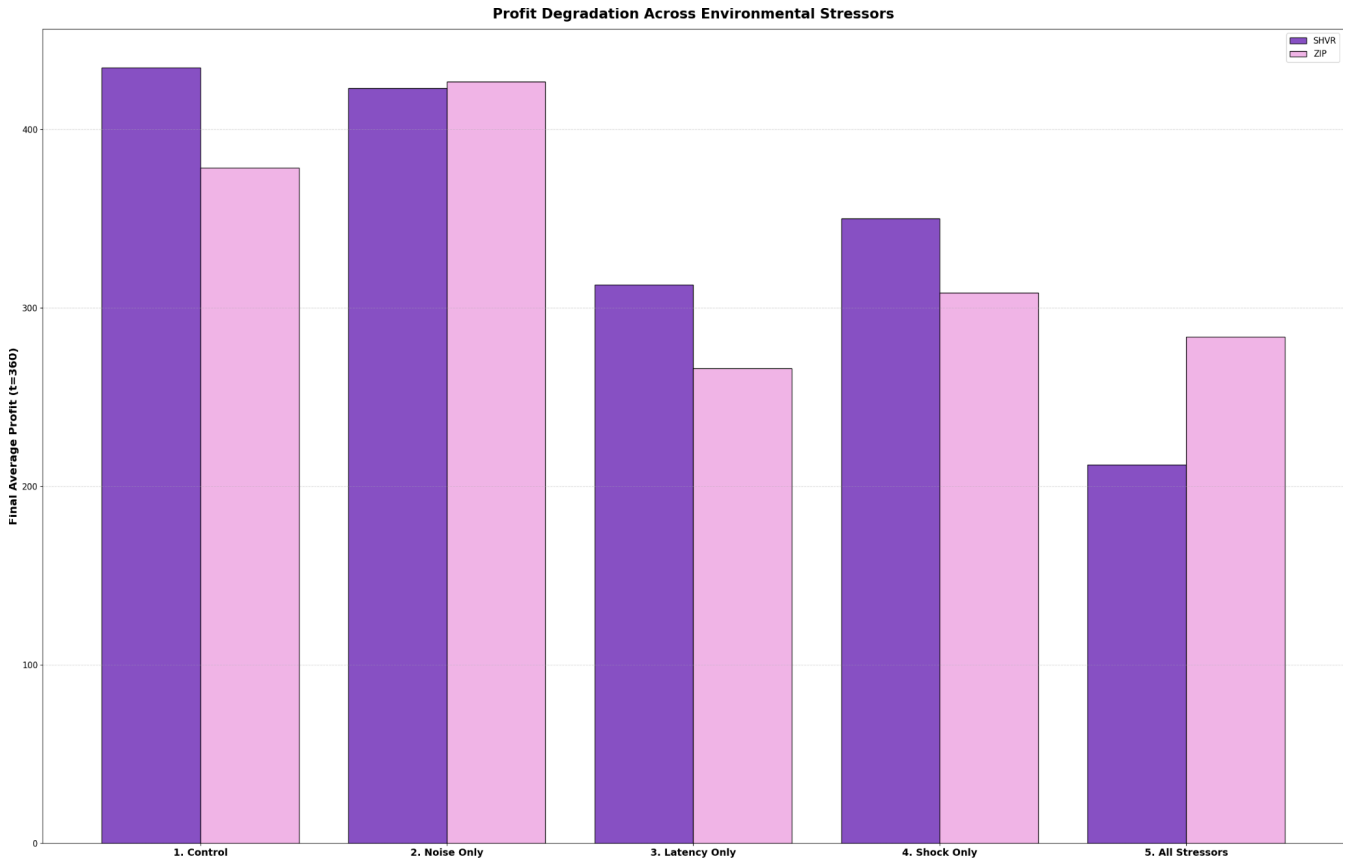


Figure 16. *Test 2.2b Profit Degradation Bar Chart Results*

Agent Efficiency Percentage (Against Insider upper bound)

Agent	1. Control	2. Noise Only	3. Latency Only	4. Shock Only	5. All Stressors
SHVR	100.00%	97.38%	72.02%	80.59%	48.81%
ZIP	87.12%	98.23%	61.20%	70.97%	65.33%

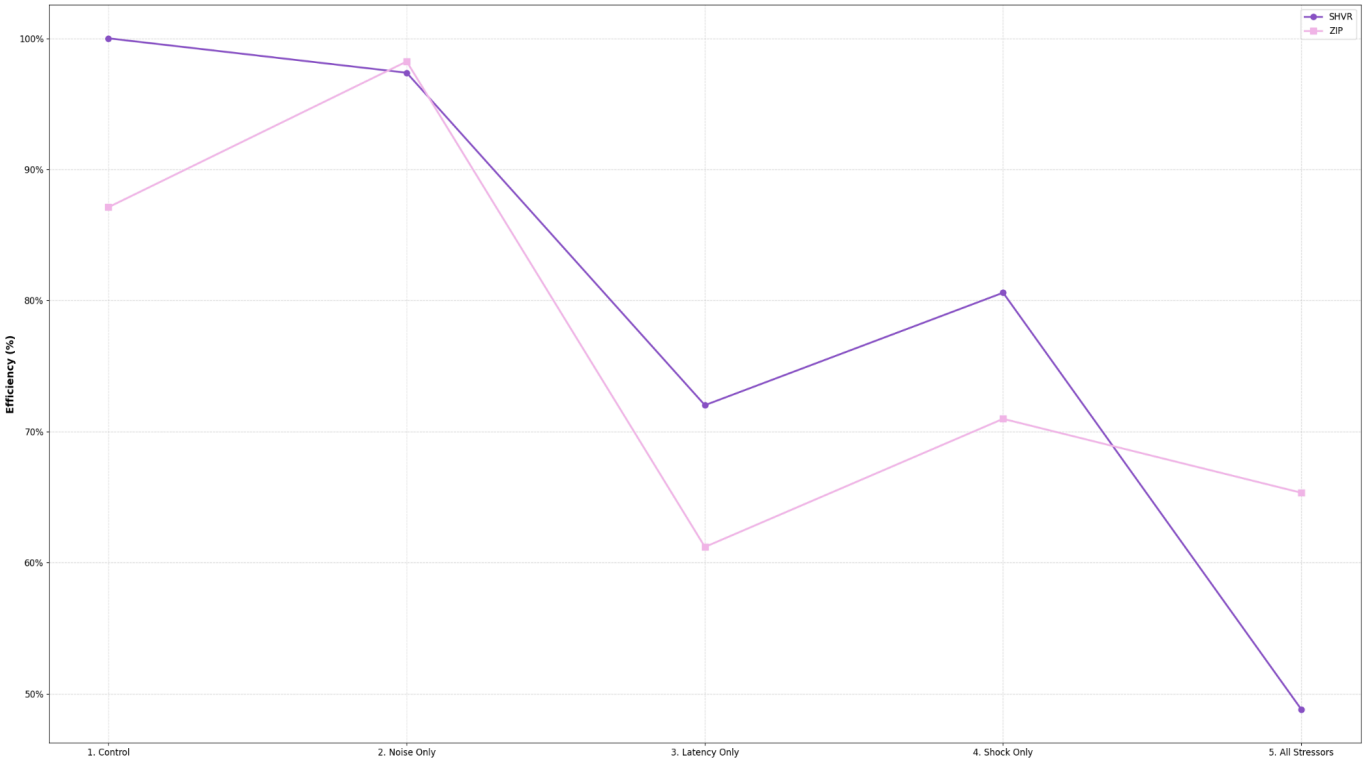


Figure 17. *Test 2.2b Market Efficiency Matrix Results*

As also seen in Test 1.2b, SHVR outperforms ZIP in the *Control* scenario, with SHVR acting as the 100% relative efficiency baseline and ZIP coming not far behind with 87.12% efficiency, reflecting Phase 1's baseline test results.

Unlike the other Phase 2 tests, *Noise Only* does not boost performance for SHVR, falling slightly to 97.38% relative efficiency while ZIP does indeed rise to 98.23%. *Latency Only* again is the hardest hitting factor, degrading both SHVR and ZIP to 72.02% and 61.20% efficiency in this scenario. Under the compounding effects of *All Stressors*, SHVR's population dominance crumbles, falling to 48.81% efficiency. However, the small population of 10 ZIP agents unexpectedly retained a moderately satisfactory performance, reclaiming market dominance at 65.33% efficiency, reestablishing itself as the more robust architecture under unfavourable conditions.

Conclusion

This project set out to investigate the research question: *How does varying market composition affect baseline market efficiency, and to what extent do compounding environmental stressors disrupt this equilibrium across distinct classes of algorithmic traders?* Through the use of the modified CDA simulator, BSE, and a custom analytical dashboard, the observed data provides a definitive answer for both parts of the question.

Phase 1 demonstrated that in an idealised environment, market composition fundamentally dictates which trading architecture achieves the greatest results. In a perfectly balanced market, adaptive margin learning (ZIP), spread manipulation (SHVR), and trend following (*Momentum*) all performed competitively, with SHVR surprisingly achieving the highest mean profit, even above Insider. However, SHVR's advantage over Insider was not statistically significant, therefore Insider was still treated as the upper bound benchmark.

Altering the population revealed many unfair dynamics. When markets were saturated with random constrained traders (ZIC) or trend-following algorithms (*Momentum*), the ZIP architecture exploited the other traders and leveraged both the weaker trend-following behaviour of *Momentum*, and the random liquidity of ZIC. However, when the market was flooded with SHVR agents, ZIP's adaptive learning methods were suppressed by SHVR's sheer volume. Additionally, the consistent failure of PRZI across all simulations confirmed that epoch-based stochastic optimisation is too slow to compete with other architectures, with the same being said for ZIC's performance.

Phase 2 answered the second half of the research question, suggesting that theoretical performances established in ideal market environments are highly fragile when subjected to real-world factors.

Signal Noise, surprisingly, generated a "boost" for several agents, exceeding their 100% control baselines when introduced. This may suggest that distorted information created exploitable price inconsistencies rather than simply reducing performance.

Execution Latency served as the most impactful factor, significantly reducing the relative efficiency of many agents. By forcing agents to forfeit execution cycles, latency exposed the dependence of many strategies on frequent market engagement. However, SHVR remained relatively strong under latency, suggesting that its spread-shaving could still extract profit when some execution opportunities were lost.

Market shocks exposed the weakness of Momentum's trend following, as sudden repricing made its historical moving averages less reliable. This reduced its ability to respond effectively after a crash, highlighting the risk of relying on price history in unstable markets.

The results demonstrate that market composition determines which agents may dominate in ideal conditions, but environmental stressors reveal whether that dominance is robust. The market equilibrium expectedly degraded significantly, causing poorer architectures like ZIC and PRZI to collapse entirely. SHVR was highly effective in clean markets and under specific stressors, but its dominance weakens under the combined effect of all environmental factors. ZIP does not always achieve the highest clean market profit, but remains one of the most resilient agents under unfavourable conditions. More broadly, this reinforces a key principle in autonomous agent design: an agent that performs well in a simplified environment may not remain effective when perception, timing, and environmental stability are disrupted. To conclude, this research demonstrates that SHVR can achieve very strong performances in ideal CDA markets, while ZIP offers much stronger resilience when market conditions become unstable.

Reflection & Future Work

Project Successes

A primary success was extending an already established system (BSE) into my own work. Shifting the base CLI to an interactive GUI was a challenge but proved valuable. Automating the entire data pipeline (running batch trials, parsing CSV outputs, and rendering visualisations) allowed for consistent high-volume testing.

The project succeeded in moving beyond the superficial profit comparisons often seen in many other projects. By implementing various statistical metrics, the analysis of data was able to explain why specific algorithms succeed or fail, not just state it.

Additionally, factoring in real-world causes of market impact allowed the project to create the link between ideal algorithmic performance and real-world financial impacts.

Project Limitations

Despite many successes, the project's scope required certain abstractions. Firstly, while the programmed environmental stressors were effective in establishing agent robustness, they were relatively binary in their implementation, and could have been adapted further to more closely represent the real world, given no time/scope constraints.

Secondly, the base BSE system is constrained to evaluating a single type of asset at any given time. Real-world financial markets are highly interconnected, where algorithmic behaviour is often dictated by multi-asset portfolios.

Finally, the balance between simulation depth and efficiency: while 30 trials of 360 second intervals provided robust statistical data for analysis, it was not long enough to observe prolonged impacts of stressors. Extending the simulation runtime to capture these effects would require an exponentially larger amount of resources.

Future Adaptations

Future iterations would focus on implementing Deep Reinforcement Learning, upgrading the logic of Momentum with Deep Q-Networks to allow agents to continuously evaluate irregular market states.

Upgrading BSE to support multi-asset CDA markets would provide an even more accurate representation of modern markets, allowing for testing of much more complex strategies such as Pairs Trading.

AI Statement

The core implementation and theory of this project was developed through my own effort, with AI being used to assist these specific sections:

- Explained the architecture and relevant functions of the base `BSE.py` code before adding my own implementations.
- Helped debug logic errors when extending the base `Trader` superclass, specifically regarding the Momentum agent's memory arrays.
- Assisted with formatting panels to ensure perfect alignment for the UI.

Additionally, no part of this report's explanations or methodology was generated by AI. It was simply used for:

- Checking grammatical and spelling errors.
- Formatting the list of citations in the References section into Harvard referencing style.

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